Antimicrobial Resistance Dashboard – *Musca domestica* Study

Area All | Organism Name All | Zone Type All | Season All

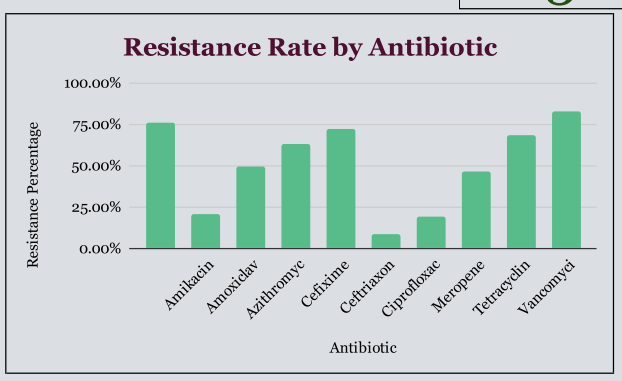
Total Isolates: 140

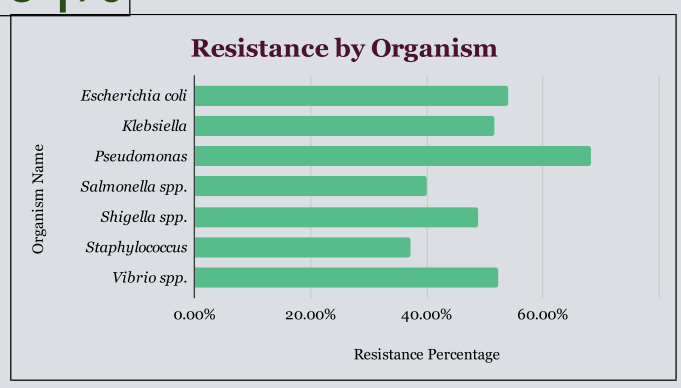
Average Resistance: 50.64%

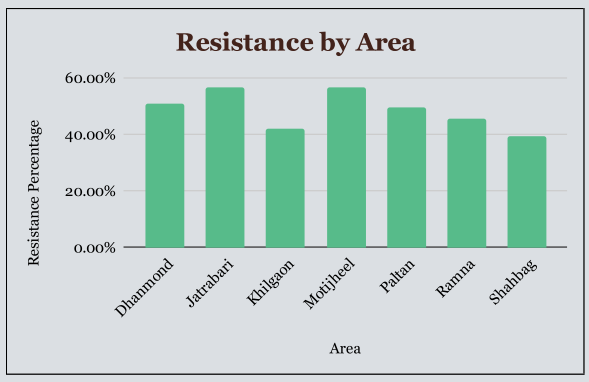
Most Isolated Bacteria: *Vibrio spp.*

Most Resistant Bacteria: *P. aureginosa*

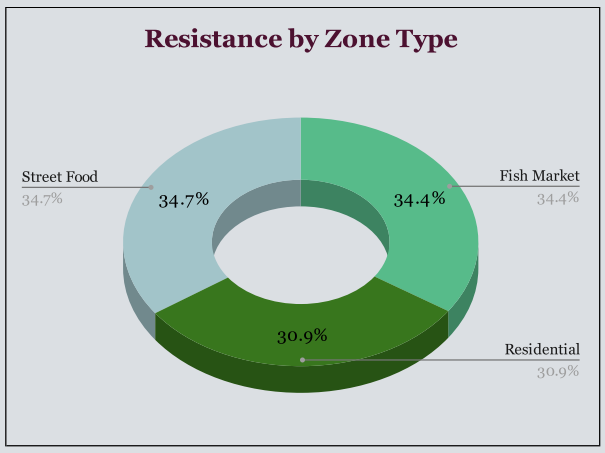
MAIN INSIGHTS

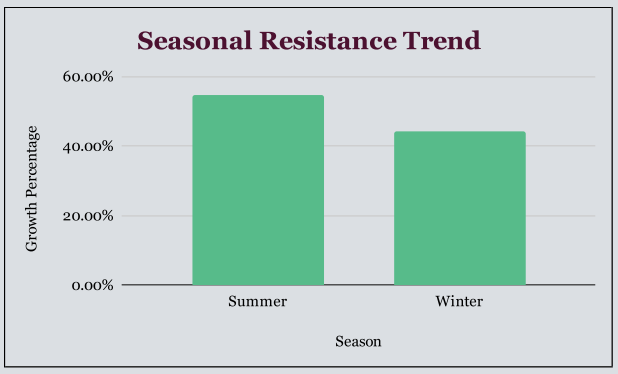


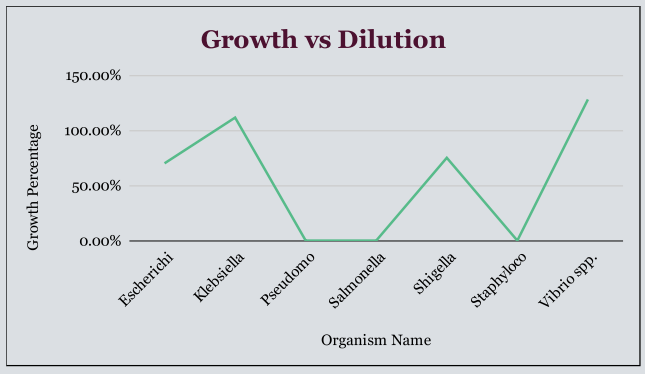




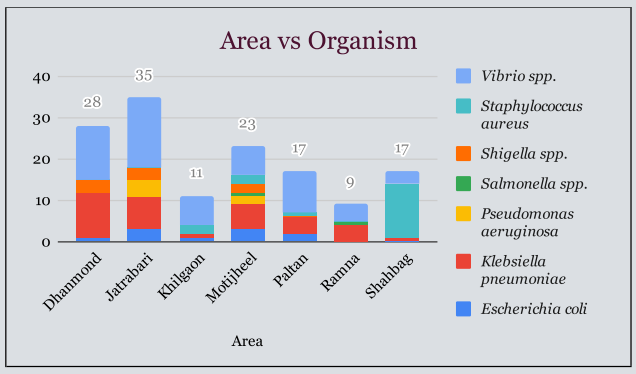
ENVIROMENT

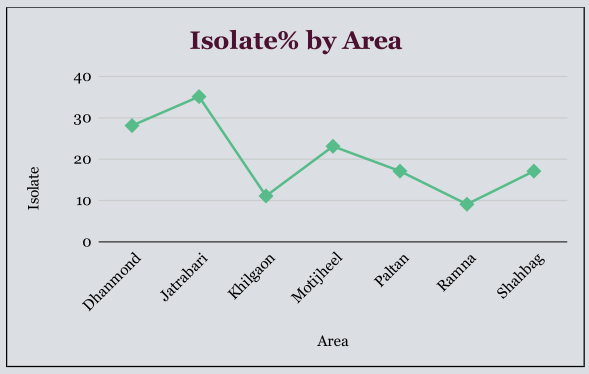






AREA DISTRIBUTION





RESISTANCE HEATMAP

HEATMAP	Organism Name						
Antibiotic	Escherichia coli	Klebsiella pneumoniae	Pseudomonas aeruginosa	Salmonella spp.	Shigella spp.	Staphylococcus aureus	Vibrio spp.
Amoxicillin	80.00%	91.43%	100.00%	0.00%	87.50%	38.89%	75.41%
Amikacin	10.00%	17.14%	16.67%	0.00%	0.00%	33.33%	24.59%
Amoxiclav	70.00%	34.29%	83.33%	0.00%	25.00%	16.67%	65.57%
Azithromycin	80.00%	65.71%	50.00%	100.00%	75.00%	44.44%	62.30%
Cefixime	80.00%	77.14%	100.00%	0.00%	62.50%	72.22%	68.85%
Ceftriaxone	0.00%	0.00%	50.00%	0.00%	0.00%	27.78%	6.56%
Ciprofloxacin	20.00%	22.86%	50.00%	0.00%	12.50%	27.78%	13.11%
Meropenem	30.00%	37.14%	83.33%	100.00%	25.00%	50.00%	50.82%
Tetracycline	70.00%	77.14%	50.00%	100.00%	100.00%	50.00%	65.57%
Vancomycin	100.00%	94.29%	100.00%	100.00%	100.00%	11.11%	90.16%

KEY INSIGHTS

1. High resistance observed in Amoxicillin and Vancomycin, indicating these antibiotics are becoming ineffective against a large proportion of isolates.
2. Ceftriaxone and Amikacin show the highest sensitivity, suggesting they remain reliable treatment options.
3. Pseudomonas aeruginosa and Vibrio spp. demonstrate elevated resistance patterns, highlighting significant risk.
4. Motijheel and Jatrabari areas show comparatively higher resistance levels, indicating possible environmental or sanitation related factors influencing bacterial spread.
5. Seasonal variation is evident, with higher resistance trends during summer, suggesting environmental conditions may influence bacterial growth and resistance.
6. Fish market and street food zones exhibit higher contamination levels, emphasizing public health concerns in food handling environments.
7. Higher dilution levels correspond to increased bacterial load, indicating variation in contamination intensity across samples